

1 Fig. 1.1 is a diagram of part of the inner membrane of a mitochondrion.

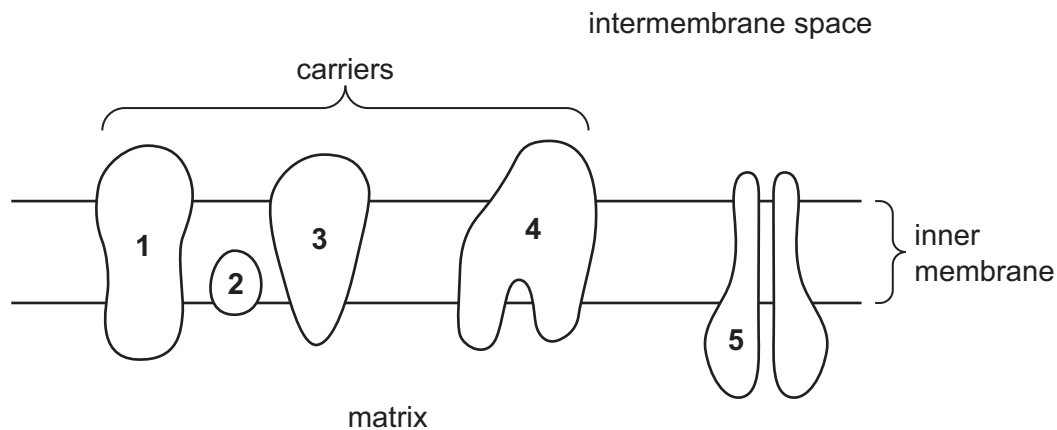


Fig. 1.1

- (a) (i) The coenzymes NAD and FAD deliver hydrogen atoms to the electron transport chain (ETC).

With reference to Fig. 1.1, state which of the four carriers receives hydrogen atoms from reduced FAD.

..... [1]

- (ii) Hydrogen atoms that are delivered to the ETC by reduced NAD and reduced FAD split into protons and electrons. Energy is released as electrons pass along the ETC.

Describe the events that occur as a result of this release of energy.

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..... [2]

- (iii) Describe how structure 5 in Fig. 1.1 is involved in oxidative phosphorylation.

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..... [3]



(b) Cyanide ions (CN^-) are highly toxic.

Cyanide ions bind to carrier **4** in Fig. 1.1 and inactivate the carrier.

Suggest **and** explain how the binding of cyanide ions to carrier **4** can have an effect on respiration.

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..... [4]

[Total: 10]